A Ray-Traced 5G Digital Twin of the IIT Gandhinagar Campus

Smrutee Behera
Roll No.25250041

Anuj Tiwari
Roll No.26310013

I. Problem Statement

5G RAN Digital Twin for Channel-Aware Resource Allocation and Network What-If Analysis

Evaluating alternative channel/resource allocations directly on a physical 5G network can be expensive and difficult to reproduce because performance depends on UE locations, propagation conditions and traffic load. A digital twin can provide a physics-aware representation of the radio environment and allow alternative allocations and network configurations to be evaluated before applying them to the network [1].

1) Deploy or use any of the digital-twin framework for eg. NVIDIA AODT and generate a scenario containing a RUs/gNBs and UEs and understand the available radio/environment configuration parameters.
2) Obtain site-specific channel information and relevant RAN measurements for different UE locations and links.
3) Define a simple baseline resource/channel-allocation strategy, such as fixed or random allocation.
4) Implement a channel-aware allocation strategy that uses digital-twin-derived channel information to select an appropriate channel/resource or serving radio for each UE.
5) Move or reconfigure UEs and evaluate how changes in the physical environment affect received signal/channel conditions and the selected allocation.
6) Compare baseline and twin-assisted strategies using relevant 5G KPIs such as SINR/channel quality, throughput, interference and resource utilization where available.
7) Use the same digital twin for a small what-if analysis, such as increasing UE density, changing UE positions or modifying resource allocation, and predict the resulting network performance before applying the configuration.
8) Document the capabilities, computational requirements and limitations of the chosen digital-twin framework as a 5G/6G network digital-twin platform and identify the research problems for which it is suitable.

II. What We Implemented

To address the project requirements, we built a ray-traced digital twin of the IIT Gandhinagar campus. A digital twin is a software model of a real place or system. In our project, it represents the campus buildings and the 5G radio network placed around them. We use this model to estimate how radio signals travel, compare different ways of assigning radio resources, and study what happens when users move or the network configuration changes.

Our scenario contains three 5G sites, with three sectors at each site, giving nine sectors in total. The network operates at 3.5 GHz. The campus model contains 74 buildings obtained from OpenStreetMap [7].

We did not connect to a live 5G network. Instead, we used NVIDIA Sionna RT [2] to simulate radio propagation in the campus environment. The implementation has the following parts.

A. Campus and 5G Network Setup

We first created a 3D model of the IIT Gandhinagar campus and placed the 5G sites and sectors in it. The campus buildings provide the physical environment through which radio signals travel.

The main settings are:

- 3 5G sites, with 3 sectors at each site.
- 9 sectors in total, operating at 3.5 GHz.
- 74 campus buildings from OpenStreetMap.
- 100 user equipments (UEs) in each random user placement used for the experiments.

B. Radio Signal and Channel Measurements

Next, we used Sionna RT to estimate how radio signals travel between each 5G sector and each user. We generated 20 different random placements, with 100 users in each placement.

For every user and sector pair, the simulation records:

- **RSRP:** The strength of the signal received from a particular sector.
- **SINR:** How strong the wanted signal is compared with interference from other transmissions and background noise.
- **Line of sight (LOS):** Whether there is a direct path between the sector and the user.
- **Path loss:** How much signal strength is lost as the signal travels from the sector to the user.

C. Baseline Resource Allocation Strategies

We compared our channel-aware method with two simple approaches. In all three strategies, each user is assumed to need 12 physical resource blocks (PRBs). A PRB is a small unit of radio resources used to transmit data.

- **B1: Nearest site with fixed PRBs.** Each user connects to the nearest 5G site. The same fixed PRB block is assigned in every sector.
- **B2: Nearest site with random PRBs.** Each user still connects to the nearest site, but the PRBs are selected randomly.

These methods act as baselines. They help us check whether using actual radio channel information from the digital twin improves the allocation.

D. Channel-Aware Allocation Using the Digital Twin

We then implemented a strategy that uses the signal measurements obtained from ray tracing.

First, for each user, the strategy checks the signal received from all available sectors and selects the sector with the strongest signal. It then estimates how strongly the sectors interfere with one another. Sectors that are likely to cause strong interference are assigned different PRBs where possible.

In simple terms, instead of connecting a user to a site only because it is geographically nearest, the strategy uses the simulated radio conditions to select a better serving sector and coordinate radio resources between sectors.

E. Comparing Network Performance

We evaluated all three strategies using the same 20 random placements of 100 users. We compared them using the following measures:

- **Average user throughput:** The average amount of data received by a user per second. A higher value means faster data transfer on average.
- **Throughput for the bottom 5% of users:** We arrange users from the lowest throughput to the highest. This value shows the throughput near the bottom of that list and helps us understand how users with the poorest connections are performing.
- **Average SINR:** Indicates how strong the wanted signal is compared with interference and noise. A higher value generally means better signal quality.
- **Average interference:** Measures the unwanted radio signal received by users. A lower interference power is better; in dBm, a more negative value means less interference.
- **PRB use:** Shows how much of the available radio resource capacity is being used.

F. Moving Users and Handover

We also simulated a user walking along three routes across the campus. As the user moves, the radio signal from the available sectors changes. The simulation checks whether the user should continue using the current sector or switch to another one.

A handover is triggered when another sector's signal is more than 3 dB stronger for two consecutive steps. When the serving gNB changes, the model shows an Xn handover. Xn is the interface that allows two neighbouring gNBs to exchange information and coordinate the transfer of a user's connection.

For each route, we record the number of handovers and the user's throughput. We also measure how often the user receives zero throughput during the walk. This lets us compare how well B1 and the Twin strategy keep a moving user connected.

G. 5G Core Network Procedures

The project also includes a message-by-message demonstration of selected 5G core network procedures. In the interactive demo, a phone moves along a campus route while the model displays the network messages associated with its connection.

The demo includes:

- **Registration:** The steps through which a phone registers with the 5G network.
- **PDU session establishment:** The steps used to set up a data connection for the phone.
- **Xn handover:** The transfer of the phone's connection from one gNB to another. Xn is the interface used by the two gNBs to coordinate this transfer.
- **Paging:** How the network tries to contact a phone when it is idle.

This is a message-level demonstration. It does not send real 5G protocol packets.

5G Core Explorer. We also built a separate 5G Core Explorer page, opened from the campus twin, which shows what happens inside the 5G core. It uses the same carrier settings as the twin (3.5 GHz, 100 MHz, 273 PRBs, 30 kHz spacing). It has two main views:

- **Network map:** The UE, two gNBs and the core network functions (AMF, SMF, UPF, AUSF, UDM, UDR, PCF, NRF, NSSF, NEF and AF) with their interfaces, such as Uu, N2, N3, N4, N6, Xn and the service-based interface. Clicking a box or an interface label explains what it does.
- **Call flow and decodes:** The same messages as a sequence chart, coloured by protocol (RRC, NAS, NGAP, XnAP, PFCP, HTTP/2 and user data). Clicking a message shows its fields one by one.

The full journey has 93 messages in nine stages: network start-up, initial access, registration, PDU session, user data, a QoS change through the NEF, handover, idle mode and paging, and deregistration. A side panel shows the phone's current state, such as its registration state, C-RNTI, 5G-GUTI, PDU session, IP address and tunnel IDs. The explorer also includes an SBI API builder, an air-interface view and a glossary. Its design is inspired by educational 5G tools such as Mpirical NetX [8].

H. What-If Analysis

We used the digital twin to estimate how network performance could change under three different conditions:

1) **Twice as many users:** Increase the number of users to study the effect of higher network load.
2) **Site B switched off:** Study how the network performs when one of the three sites is unavailable.
3) **Users moved by about 30 m:** Change user positions and study how the radio conditions and allocation change.

For each case, the twin first predicts the network performance. We then run the changed scenario with additional random signal variation and compare the result with the prediction. In this experiment, the additional variation is used to represent random shadowing. It is a simulation, not a measurement from a real 5G network.

I. Computational Requirements and Limitations

We also measured how much time and memory the ray-tracing process requires. The experiments were run on an Apple M4 laptop using its CPU, without an NVIDIA GPU.

The main software tools are Sionna RT for ray tracing, Mitsuba and Dr.Jit for the rendering and computation used by Sionna, Python and NumPy for the simulation, Matplotlib for plots, and three.js for the interactive 3D campus page. OpenStreetMap provides the campus building data.

III. Architecture and Design

The system has two main parts. The **radio twin** estimates radio signals using the campus layout. The **network layer** uses these estimates to assign radio resources and display the 5G procedures.

Radio twin. The campus building outlines from OpenStreetMap are converted into 3D blocks. Sionna RT traces radio paths from each of the nine sectors to each user. The simulation considers up to three reflections and diffraction around building edges. The resulting signal information is used to calculate RSRP, SINR and line of sight.

Resource allocation. All three strategies use the same traffic assumption: each user needs 12 PRBs. They also use the same KPI calculations, so the main difference is how they choose the serving sector and PRBs:

- **B1:** Each user connects to the nearest site. Every sector uses a fixed PRB block starting at PRB 0, so neighbouring sectors overlap.
- **B2:** Each user connects to the nearest site, and PRBs are selected randomly in each sector.
- **Twin:** Each user connects to the sector with the strongest ray-traced signal. The strategy estimates interference between sector pairs, processes the busiest sectors first, and gives each sector the PRBs that are least likely to overlap with its strongest interferers.

Mobility and 5G core. As a user walks, the serving sector changes when another sector is more than 3 dB stronger for two consecutive steps. This is the A3 handover condition used in the simulation. A change of site starts an Xn handover between gNBs. The core model displays registration through the AMF, AUSF and UDM; PDU session establishment through the AMF, SMF and UPF; data transfer through the N3 tunnel; and paging of an idle phone across its tracking area list (TAI list) [5], [6].

Design choices. Table I summarizes the main implementation choices.

TABLE I
Main design choices

Choice	Why	Trade-off
Sionna RT, not AODT [4]	Runs on a laptop CPU; AODT needs NVIDIA GPUs and Docker	Models the radio channel, not the complete network
Masts beside buildings, 30 m	Used because 60 of 65 users within 100 m of a mast had no line of sight when the masts were placed on roofs	Requires separate towers
One value per link	Keeps the channel model simple to explain and use	Does not show frequency-dependent effects
Greedy PRB rule	Gives a simple allocation method that runs in under 1 s	Not guaranteed to be optimal; some edge users can lose out
Message-level core model	Uses the twin's handovers and shows the procedures taught in class	Does not create real NGAP/PFCP packets

IV. Implementation

The project is written in Python. Sionna RT performs the ray tracing, NumPy is used for calculations, and Matplotlib generates the result plots. The interactive demonstration is an offline HTML page that uses three.js to display the 3D campus.

The code is organized into the radio twin (`twin/`), the experiment scripts (`tasks/`), the 5G network model (`net/`), and the demo files (`run.py` and `viewer.html`). The test file `test_flows.py` checks that the displayed network procedures follow the expected order.

The Twin strategy assigns resources in two steps. First, it connects each user to the sector with the strongest simulated signal. Next, it checks which sectors are likely to interfere with one another and tries to assign them different PRBs. This reduces overlap between sectors that may cause strong interference. The user's throughput is then estimated from its SINR on its assigned PRBs, using a 10% block-error target and 14% overhead. Most users get 12 PRBs; about 5% of users get fewer, because a sector with more than 22 users runs out of its 273 PRBs. The model limits one user's throughput to 23.1 Mbps.

Challenges and fixes.

- *Fixed and random PRBs initially gave the same result.* When every sector uses the whole band, changing the PRB assignment does not change which sectors overlap. We therefore used a partial-load model in which each user needs 12 PRBs. Neighbouring sectors then interfere when their assigned PRBs overlap.
- *Very weak links affected an early result.* In an earlier version of the code, very weak ray-traced links made the baseline look worse than it should. Replacing them with values from the 3GPP model reduced the gain reported at that time from 137% to 129%. In the current code, we keep the ray-traced values as they are, and when calculating the average SINR we count any user below -20 dB as -20 dB.

TABLE II
TESTBED AND SCENARIO SETTINGS

Item	Setting
Hardware	Apple M4, 10 CPU cores, 16 GB RAM, no GPU
Software	macOS 26.5.2, Python 3.14.7, sionna-rt 2.2.0, mitsuba 3.9.1, drjit 1.5.0 (CPU), numpy 2.5.4
Scene	IITGN, 1520 m × 1520 m, 74 buildings (3768 triangles), tallest 24 m
Sites	A, B, C; 30 m masts; 3 sectors each; 8° downtilt
Radio	3.5 GHz (n78), 100 MHz, 30 kHz spacing, 273 PRBs, 46 dBm per sector, 3GPP sector antenna [3]
Users	1.5 m high, 9 dB noise figure; 100 users per set, 20 sets
Ray tracing	10^6 rays per sector, up to 3 reflections, diffraction on
Mobility	3 routes, 66 points 2 m apart
Core	1 UE, 3 gNBs, AMF, AUSF, UDM/SIDF, NSSF, SMF, UPF; test IMSI 001-01-9876500001, slice SST 1, 5QI 9

- *The walking user was initially always last in the user list.* This could give B1 an unfair advantage. We now place the walking user at a random position in the list and average the results over 10 user sets.
- *Results vary slightly between runs.* Sionna RT on a CPU is not exactly repeatable, and laptop performance can decrease as it heats up. We report averages with 95% confidence intervals and use the median of three timing runs.

Code: https://github.com/smrutee20/5G-digital-twin

V. TESTBED SETUP

All experiments were run on one laptop using the CPU only. The main settings are listed in Table II. Rebuilding the demo from saved data takes about 1 s. The full study takes about 15 minutes when the ray-tracing results are already saved, and about 25 minutes when run from scratch.

VI. EVALUATION

A. Radio channel measurements

For one set of 100 users, the simulation traces 900 user–sector links in 17.6 s. Of these links, 61% have a direct line of sight. For 75% of the users, the link to their strongest sector has a direct line of sight. The median signal strength from the strongest sector is -75.2 dBm.

The nearest site is not always the one providing the strongest signal. The Twin strategy moves 44 of the 100 users to a different sector. For these users, the selected sector is stronger by a median of 8.8 dB.

B. Comparing the allocation strategies

Table III compares the three strategies across 20 sets of 100 users.

The Twin strategy gives the highest average user throughput: 10.92 Mbps. This is 69% higher than B1 and 26% higher than B2. It also gives the best result for the users with the poorest connections. The throughput for the bottom 5% of users is 1.33 Mbps with the Twin strategy, compared with 0 Mbps for B1 and 0.05 Mbps for B2.

TABLE III
NETWORK PERFORMANCE FOR THE THREE STRATEGIES (20 SETS OF 100 USERS; MEAN ± 95% CONFIDENCE INTERVAL WHERE AVAILABLE)

Performance measure	B1 fixed	B2 random	Twin
Average user throughput (Mbps)	6.48±0.50	8.67±0.35	**10.92±0.50**
Throughput for the bottom 5% of users (Mbps)	0.00±0.00	0.05±0.05	**1.33±0.29**
Average SINR (dB)	4.85±0.81	**15.12±0.81**	13.85±0.66
Average interference per PRB (dBm)	−66.84	−69.57	**−71.98**
PRB use (%)	48.74	48.74	48.49

TABLE IV
PERFORMANCE OF THE WALKING USER ON EACH ROUTE (AVG. THROUGHPUT / % OF STEPS AT 0 MBPS)

Route	Handovers	B1	Twin
1	None	0.5 Mbps / 76%	**18.8 Mbps / 0%**
2	2 (56, 92 m)	2.8 Mbps / 50%	2.7 Mbps / **0%**
3	1 (64 m)	**6.4 Mbps** / 50%	3.9 Mbps / **6.5%**

The average interference is also lowest with the Twin strategy. Its value is -71.98 dBm per PRB. Since a more negative dBm value means lower interference power, this is better than the values for B1 and B2.

B2 has a slightly higher average SINR than the Twin strategy. This happens because each user's SINR is an average over its PRBs. With random PRBs, some of a user's PRBs are almost free of interference, while others are heavily interfered. The clean PRBs push the average SINR up, but the interfered PRBs carry little data. At its average SINR, a B2 user would get about 4.3 Mbps more than it actually does; for the Twin strategy, the difference is only 0.3 Mbps.

We also checked the effect of the two main parts of the Twin strategy separately using one set of users. Selecting the strongest serving sector increases average throughput from 4.29 to 5.93 Mbps. PRB coordination alone gives 7.55 Mbps, while using both together gives 8.55 Mbps.

C. Users moving across the campus

We simulated one user walking along each of the three routes while 100 other users remained in place. The results are averaged over 10 sets of users.

The table shows two things: the walking user's average throughput, and the percentage of steps during which the user receives no data. A value of 0% does not mean that the throughput is zero. It means that the user received some data at every step. For example, on Route 1, B1 gives the user no data in 76% of the steps. With the Twin strategy, the user receives data throughout the route. On Route 2, the Twin strategy also has 0% zero-throughput steps, even though its average throughput (2.7 Mbps) is slightly lower than B1 (2.8 Mbps).

The Twin strategy avoids zero-throughput steps on Routes 1 and 2, and has only 6.5% on Route 3. However, its average throughput is not higher on every route. On Route 3, B1 gives

TABLE V
PREDICTED AND SIMULATED RESULTS FOR THE WHAT-IF SCENARIOS

Network condition	Avg. throughput (Mbps) Predicted / simulated	Avg. SINR (dB) Predicted / simulated
Current network	8.55 / 9.39	10.71 / 13.36
Twice as many users	4.72 / 4.77	6.68 / 6.54
Site B switched off	7.80 / 8.26	10.21 / 11.32
Users moved by $\approx$ 30 m	7.11 / 7.97	9.19 / 10.60

6.4 Mbps while the Twin strategy gives 3.9 Mbps. In this case, during the first half of the route, the fixed PRBs used by B1 happen to avoid some neighbouring transmissions. The Twin strategy chooses PRBs to protect the sector's users overall, but the walking user still experiences interference from a strong neighbouring sector.

These results show why it is useful to look at both average throughput and the time a user receives zero throughput.

D. What-if analysis

We used the digital twin to predict the effect of three network changes. The predicted values are compared with a second simulation of each changed scenario. In the second simulation, each link receives an additional random signal variation of 4 dB. The table labels this second simulation as "simulated". It is not a measurement from a physical 5G network.

For all four cases in the table, the predicted average throughput is within 0.9 Mbps of the result from the second simulation. The predicted SINR is within 2.7 dB. The predictions also show the same overall direction of change as the second simulation.

For example, doubling the number of users reduces the average throughput per user to roughly half. Switching off Site B reduces throughput by a smaller amount, because Sites A and B cover some of the same area.

E. Digital-twin platform: capabilities, cost and limitations

Capabilities. Sionna RT calculates direct, reflected and diffracted radio paths between the sectors and users. Antenna settings, transmission power, tilt and building materials can be changed to study different radio conditions.

Computational cost. Loading the scene takes 0.4–0.6 s. At 10^6 rays, tracing takes about 20–32 ms per user–sector link: 17.6 s for 900 links and 117 s for 3600 links. The time increases with the number of rays: for 900 links, it takes 5.6 s at 2×10^5 rays and 78 s at 3×10^6 rays. Memory use is 0.7–1.1 GB. The laptop became up to 50% slower as it heated during longer runs.

Limitations. The implementation models the radio channel, but it does not include a complete RAN stack or a real 5G core. The campus scene is static and does not include trees, even though parts of the campus are wooded. Some buildings use assumed heights because height information is missing from OpenStreetMap. The twin is also not updated using measurements from a real network.

Suitable research problems. This type of digital twin is useful for studying where to place sites and sectors before deployment, reducing interference between sectors, planning handovers and tracking areas, and estimating the effect of switching off a site to save energy. It is not designed to study core-network signalling load or perform real-time network control. Those tasks would require a more complete platform, such as AODT [4], connected to a real 5G core.

VII. SUMMARY

We built a ray-traced digital twin of the IIT Gandhinagar campus to study 5G radio resource allocation and network changes in a simulated environment. The twin uses the campus building layout to estimate radio conditions between users and the nine sectors at three 5G sites.

Across 20 sets of 100 users, the Twin strategy increased average user throughput from 6.48 to 10.92 Mbps compared with the fixed-allocation baseline. It also improved throughput for the bottom 5% of users from 0 to 1.33 Mbps and reduced average interference. In the mobility experiments, the Twin strategy kept the walking user connected for most of the simulated routes. The what-if predictions were within 0.9 Mbps of the results from the second simulations.

The project also provides an interactive 3D campus page that displays a moving user and message-level demonstrations of selected 5G core procedures.

Limitations: The campus scene does not include trees and uses assumed heights for some buildings. Each link is represented by one signal value, and the what-if comparison uses another simulation rather than a real network. The PRB allocation is a simple greedy method and can still give poor results to some users, as seen on Route 3. The 5G core procedures are shown as messages with assumed delays, not as real protocol packets.

Future work: Future improvements include adding trees and measured building heights, updating the twin using measurements from a real network, improving resource allocation for users at the edge of a cell or about to hand over, testing changes such as antenna downtilt, and running the study on a GPU or on AODT with a real 5G core such as Open5GS.

VIII. REPRODUCIBILITY

The code, data and instructions are available at https://github.com/smrutee20/5G-digital-twin. After installing the packages listed in `requirements.txt`, run `python run.py` to rebuild the demo page `twin.html`. The scripts in `tasks/` can be used to run the experiments again.

REFERENCES

[1] 3GPP, "Study on management aspects of Network Digital Twin," TR 28.915, Release 19. [Online]. Available: https://www.etsi.org/deliver/etsi_tr/128900_128999/128915/19.00.00_60/tr_128915v190000p.pdf

[2] J. Hoydis, F. Aït Aoudia, S. Cammerer, M. Nimier-David, N. Binder, G. Marcus, and A. Keller, "Sionna RT: Differentiable ray tracing for radio propagation modeling," arXiv:2303.11103, 2023.

[3] 3GPP, "Study on channel model for frequencies from 0.5 to 100 GHz," TR 38.901.

[4] NVIDIA, "Aerial Omniverse Digital Twin (AODT)," documentation, 2024.

[5] 3GPP, “System architecture for the 5G System (5GS),” TS 23.501.
[6] 3GPP, “Procedures for the 5G System (5GS),” TS 23.502.
[7] OpenStreetMap contributors, “OpenStreetMap.” [Online]. Available: https://www.openstreetmap.org
[8] Mpirical, “NetX.” [Online]. Available: https://www.mpirical.com/netx

APPENDIX A
TEAM WORK DISTRIBUTION AND MEMBER RATING

#	Member	Tasks done	Rating
1	Smrutee Behera	Built the IITGN campus scenario in Sionna RT and collected ray-traced channel measurements. Designed and evaluated the baseline and twin-assisted allocation strategies, including the mobility study, KPI comparison and what-if analysis. Interactive 3D campus UI (demo page). Prepared the project report. Platform evaluation (jointly).	10
2	Anuj Tiwari	5G core procedures: registration, PDU session, Xn handover and paging. Linking the 5G core flow with the moving user in the demo. Built the 5G Core Explorer: network map, call flow with message decodes, SBI API builder, air-interface view and glossary. Prepared the PPT. Platform evaluation (jointly).	9